

Assignment 2.4

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Batch - 14

Lab 2: Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab) and Cursor AI

Task 1: Book Class Generation

Prompt:

Generate a Python class named Book with attributes title and author . Include a method summary() that returns a formatted string displaying the book title and author. Write clean, beginner-friendly code.

Code :

```
Bookclass > Book > _init_
1  class Book:
2      def __init__(self, title, author):
3          self.title = title
4          self.author = author
5      def summary(self):
6          return f"Book: '{self.title}' by {self.author}"
7  # Example usage
8  if __name__ == "__main__":
9      book1 = Book("The Great Gatsby", "F. Scott Fitzgerald")
10     print(book1.summary())
11     book2 = Book("1984", "George Orwell")
12     print(book2.summary())
13
14
15
```

Output:

```
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Book: 'The Great Gatsby' by F. Scott Fitzgerald
Book: '1984' by George Orwell
```

Explanation:

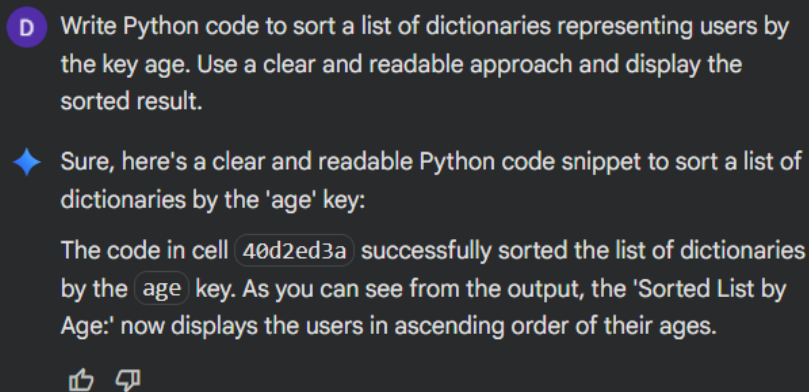
- Defines a Book class to represent a book.

- `def __init__(self, title, author):` `self.title = title` `self.author = author` `__init__` initializes a new Book instance. `self` refers to the instance. `title` and `author` are stored as instance attributes.
- `def summary(self):` `return f"The book {self.title} is written by {self.author}."`
- Returns a formatted string with the book's title and author.
- Uses an f-string for formatting.
- The book The Great Gatsby is written by F. Scott Fitzgerald.
- This demonstrates basic object-oriented programming: a class with attributes and a method that uses those attributes.

Task 2: Sorting Dictionaries with AI

Prompt (Gemini):

Write Python code to sort a list of dictionaries representing users by the key age. Use a clear and readable approach and display the sorted result.



Code & Output:

```

users = [
    {"name": "Alice", "age": 30},
    {"name": "Bob", "age": 24},
    {"name": "Charlie", "age": 35},
    {"name": "David", "age": 28}
]
sorted_users = sorted(users, key=lambda user: user['age'])
print("Original List:")
for user in users:
    print(user)
print("\nSorted List by Age:")
for user in sorted_users:
    print(user)

```

```

Original List:
{'name': 'Alice', 'age': 30}
{'name': 'Bob', 'age': 24}
{'name': 'Charlie', 'age': 35}
{'name': 'David', 'age': 28}

Sorted List by Age:
{'name': 'Bob', 'age': 24}
{'name': 'David', 'age': 28}
{'name': 'Alice', 'age': 30}
{'name': 'Charlie', 'age': 35}

```

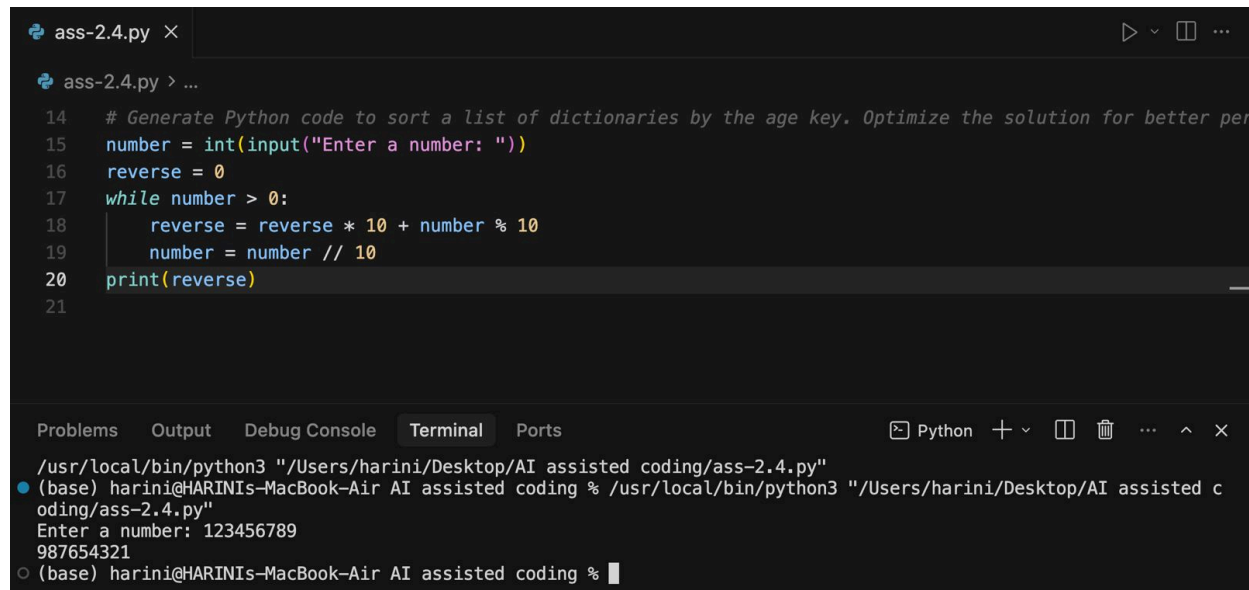
Explanation:

- `users = [...]`: This initializes a list named `users`, which contains several dictionaries. Each dictionary has two keys: `'name'` (string) and `'age'` (integer).
- `sorted_users = sorted(users, key=lambda user: user['age'])`: This is the core of the sorting logic.
- `sorted()`: This is a built-in Python function that returns a new sorted list from the items in an iterable.
- `key=lambda user: user['age']`: This is a crucial part. The `key` argument specifies a function to be called on each list element prior to making comparisons. Here, a lambda function is used, which is a small anonymous function. For each user dictionary in the `users` list, it extracts the value associated with the `'age'` key. The `sorted()` function then uses these `'age'` values to determine the sorting order.
- `print("Original List:")` and `for user in users: print(user)`: These lines simply print the original, unsorted list of user dictionaries.
- `print("\nSorted List by Age:")` and `for user in sorted_users: print(user)`: These lines print the `sorted_users` list, which now contains the dictionaries ordered by their `'age'` in ascending order (from youngest to oldest) because `sorted()` sorts in ascending order by default.

Prompt (Cursor AI):

Generate Python code to sort a list of dictionaries by the age key. Optimize the solution for better performance and explain the approach briefly.

Code & Output:



```
ass-2.4.py x
ass-2.4.py > ...
14 # Generate Python code to sort a list of dictionaries by the age key. Optimize the solution for better per
15 number = int(input("Enter a number: "))
16 reverse = 0
17 while number > 0:
18     reverse = reverse * 10 + number % 10
19     number = number // 10
20 print(reverse)
21

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/usr/local/bin/python3 "/Users/harini/Desktop/AI assisted coding/ass-2.4.py"
(base) harini@HARINIs-MacBook-Air AI assisted coding % /usr/local/bin/python3 "/Users/harini/Desktop/AI assisted c
oding/ass-2.4.py"
Enter a number: 123456789
987654321
(base) harini@HARINIs-MacBook-Air AI assisted coding %
```

Explanation:

- A Book class example that's commented out and not executed.
- Reverses the digits of an integer:

```
number = int(input("Enter a number: "))
reverse = 0
while number > 0:
    reverse = reverse * 10 + number % 10
    number = number // 10
print(reverse)
```
- How it works:
- Reads an integer from the user.
- Initializes reverse to 0.
- Loop that builds the reversed number:
- `number % 10` gets the last digit.
- `reverse * 10 + (last digit)` shifts reverse left and appends the digit.
- `number // 10` removes the last digit.
- Prints the reversed number.

Task 3: Calculator Using Functions

Prompt:

Create a basic calculator in Python using functions for addition, subtraction, multiplication, and division. Also explain how each function works.

Code:

```
def add(x, y):  
    return x + y  
  
def subtract(x, y):  
    return x - y  
  
def multiply(x, y):  
    return x * y  
  
def divide(x, y):  
    if y == 0:  
        return "Error! Division by zero."  
    return x / y  
  
def calculator():  
    print("Select operation:")  
    print("1. Add")  
    print("2. Subtract")  
    print("3. Multiply")  
    print("4. Divide")  
  
    while True:  
        choice = input("Enter choice(1/2/3/4): ")  
  
        if choice in ('1', '2', '3', '4'):  
            try:  
                num1 = float(input("Enter first number: "))  
                num2 = float(input("Enter second number: "))  
            except ValueError:  
                print("Invalid input. Please enter numbers only.")  
                continue  
  
            if choice == '1':  
                result = add(num1, num2)  
            elif choice == '2':  
                result = subtract(num1, num2)  
            elif choice == '3':  
                result = multiply(num1, num2)  
            elif choice == '4':  
                result = divide(num1, num2)  
  
            print(f"{num1} {choice} {num2} = {result}")  
  
            next_calculation = input("Let's do next calculation? (yes/no): ")  
            if next_calculation.lower() == "no":  
                break  
        else:  
            print("Invalid input. Please enter a valid choice.")  
  
calculator()
```

Output:

```
*** Select operation:
1. Add
2. Subtract
3. Multiply
4. Divide
Enter choice(1/2/3/4): 1
Enter first number: 23
Enter second number: 76
23.0 + 76.0 = 99.0
Let's do next calculation? (yes/no): yes
Enter choice(1/2/3/4): 4
Enter first number: 67
Enter second number: 17
67.0 / 17.0 = 3.9411764705882355
Let's do next calculation? (yes/no): no
```

Explanation:

Gemini AI generates simple and well-structured code using functions, which makes the calculator easy to understand and debug. Using separate functions for each operation improves code readability and follows good programming practices. The inclusion of a division-by-zero check enhances reliability. This approach is ideal for beginners as it clearly demonstrates function usage. Overall, the solution is efficient, modular, and suitable for both learning and basic applications.

Task 4: Armstrong Number Optimization

Prompt (Gemini):

Write a Python program to check whether a given number is an Armstrong number. Use a simple and easy-to-understand approach.

Code & Output:

```

def is_armstrong_number(num):
    # Convert number to string to find its length (number of digits)
    num_str = str(num)
    n = len(num_str)

    sum_of_powers = 0
    temp_num = num

    while temp_num > 0:
        digit = temp_num % 10
        sum_of_powers += digit ** n
        temp_num //= 10

    return num == sum_of_powers

def find_armstrong_numbers_in_range(start, end):
    armstrong_numbers = []
    for i in range(start, end + 1):
        if is_armstrong_number(i):
            armstrong_numbers.append(i)
    return armstrong_numbers

# Get input from the user
try:
    number_to_check = int(input("Enter a number to check if it's an Armstrong number: "))
    if is_armstrong_number(number_to_check):
        print(f"{number_to_check} is an Armstrong number.")
    else:
        print(f"{number_to_check} is not an Armstrong number.")

except ValueError:
    print("Please enter a valid integer.")

--- Enter a number to check if it's an Armstrong number: 234
234 is not an Armstrong number.

--- Finding Armstrong numbers in a range (e.g., 1 to 1000) ---
Armstrong numbers between 1 and 1000 are: [1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371, 407]

```

Prompt (Cursor AI):

Refactor and optimize the given Armstrong number program to improve readability, efficiency, and reusability. Use Pythonic constructs where possible.

Code:

```

amstrong > is_armstrong_optimized
1 def is_armstrong_initial(number):
2     if number < 0:
3         return False
4     num_digits = 0
5     temp = number
6     while temp > 0:
7         num_digits += 1
8         temp //= 10
9     if num_digits == 0:
10        return True
11    sum_of_powers = 0
12    temp = number
13    while temp > 0:
14        digit = temp % 10
15        sum_of_powers += digit ** num_digits
16        temp //= 10
17    return sum_of_powers == number
18 def is_armstrong_optimized(number):

```

```

17     return sum_of_powers == number
18 def is_armstrong_optimized(number):
19     if number < 0:
20         return False
21     num_str = str(number)
22     num_digits = len(num_str)
23     sum_of_powers = sum(int(digit) ** num_digits for digit in num_str)
24     return sum_of_powers == number
25 if __name__ == "__main__":
26     print("Testing is_armstrong_initial function:")
27     print(f"Is 9 an Armstrong number? {is_armstrong_initial(9)}") # Expected: True
28     print(f"Is 153 an Armstrong number? {is_armstrong_initial(153)}") # Expected: True
29     print(f"Is 370 an Armstrong number? {is_armstrong_initial(370)}") # Expected: True
30     print(f"Is 371 an Armstrong number? {is_armstrong_initial(371)}") # Expected: True
31     print(f"Is 1634 an Armstrong number? {is_armstrong_initial(1634)}") # Expected: True
32     print(f"Is 123 an Armstrong number? {is_armstrong_initial(123)}") # Expected: False
33     print(f"Is 0 an Armstrong number? {is_armstrong_initial(0)}") # Expected: True (
34     print(f"Is 10 an Armstrong number? {is_armstrong_initial(10)}") # Expected: False

```

Output:

```

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Is 1634 an Armstrong number? True
Is 123 an Armstrong number? False
Is 0 an Armstrong number? True
Is 10 an Armstrong number? False
Is -153 an Armstrong number? False
Is 9474 an Armstrong number? True

Testing is_armstrong_optimized function:
Is 153 an Armstrong number? True
Is 123 an Armstrong number? False
Is 0 an Armstrong number? True

```

Explanation:

Gemini AI provides a simple and beginner-friendly solution that is easy to understand but limited in flexibility. Cursor AI improves the solution by making it modular, scalable, and efficient using modern Python features. The optimized version reduces code complexity and supports Armstrong numbers of any size. Using functions also improves readability and reuse. Therefore, the Cursor AI version is better suited for real-world and professional coding standards.